Ahmad Samim Hamza

Software Engineer | Applied AI

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ABOUT ME

Software Engineer specializing in Applied AI and Master's student in Artificial Intelligence at the University of Jyväskylä, with 5 years of experience building web applications, backend APIs, and internal business systems. At Konecranes, I designed, evaluated, and deployed an internal enterprise RAG and multi-agent assistant on AWS (Bedrock, LangGraph, LangChain) behind a Next.js and FastAPI application with PostgreSQL feedback storage and CloudWatch monitoring. Earlier in my career, I led an 8-person engineering team building Dara.af and developed production ERP and logistics systems. My core stack includes Python, TypeScript, React, Next.js, FastAPI, NestJS, Laravel, PostgreSQL, and AWS.

SKILLS

Applied AI, RAG & Multi-Agent Systems: RAG | Corrective RAG (CRAG) | Multi-Agent Systems | Agentic Workflows | LangGraph | LangChain | AWS Bedrock (Agents & AgentCore) | Hybrid Search (BM25 + Dense Vectors) | Reciprocal Rank Fusion (RRF) | Cross-Encoder Reranking | Vector Stores (Qdrant, ChromaDB, S3) | LLM Evaluation

Backend & Data Engineering: Python (FastAPI, AsyncIO) | TypeScript/JavaScript (NestJS, Node.js, Express) | PostgreSQL (Schema Design, Query Optimization, pgvector) | PHP (Laravel, Filament, Livewire) | REST APIs | MySQL | SQLite | Redis

Frontend Engineering: Next.js | React | TypeScript | Tailwind CSS | Material UI | Mantine | WebSockets | Vue.js | Nuxt.js

Cloud & DevOps: AWS (Bedrock, ECS, Amplify, RDS, S3, CloudWatch) | Docker | CI/CD Pipelines (GitHub Actions) | Linux

AI-Assisted Development: Claude Code | Cursor | Antigravity

Academic Coursework & Research: Semantic Web (Ontologies, RDF, OWL, SPARQL) | PyTorch | scikit-learn | Multi-Objective Optimization | Kubernetes

WORK EXPERIENCE

Software Development Specialist (AI/ML) – Fixed-Term Summer Contract 05/2026 – 08/2026
Konecranes Hyvinkää, Finland

- Built, evaluated, and deployed an internal knowledge assistant to help engineers search across 100+ technical documents (PDF, XML, and XLSX) indexed in S3.
- Tested and compared three RAG architectures using AWS Bedrock (Agents & AgentCore), LangChain, and LangGraph, connecting them to a shared Next.js and FastAPI web application.
- Implemented a supervisor/collaborator multi-agent workflow to break down multi-step technical queries across internal knowledge sources.
- Used PostgreSQL (AWS RDS) for user authentication, chat sessions, and a human feedback pipeline so internal testers could rate and compare answers across the three RAG setups.
- Deployed the application on AWS (Amplify, ECS, RDS) and used CloudWatch logs and error tracking to troubleshoot retrieval quality and latency bottlenecks.
- Worked directly with internal domain users, wrote setup and architecture documentation, and handed the system over to the internal engineering team.

Senior Software Engineer 05/2024 – 07/2025
Ramaki Group of Companies Kabul, Afghanistan

- Led an 8-person engineering team building Dara.af, an e-commerce platform supporting hundreds of product listings and daily customer orders using Laravel, PostgreSQL, React, and Next.js.
- Designed and optimized normalized PostgreSQL database schemas for orders, inventory, and merchant catalogs.
- Developed RESTful APIs and frontend features, and wrote automated unit, integration, and feature tests to prevent checkout and stock regressions.
- Reviewed pull requests, set team coding standards, and mentored junior developers.

Software Engineer 09/2022 – 04/2024
Peace Global Logistics Kabul, Afghanistan

- Built core ERP modules for international logistics, shipment tracking, warehouse inventory, HR, and finance using NestJS, React, Next.js, and PostgreSQL.
- Designed relational database schemas around real-world shipment milestones and optimized SQL queries for operational reporting across high-volume shipment tables.
- Created reusable React and Tailwind CSS components, working closely with logistics and finance staff to replace manual spreadsheets with tested web workflows.

Instructor (Python & Machine Learning) 04/2022 – 10/2022
Kabul University Kabul, Afghanistan

- Taught introductory Python programming and machine learning basics (data preprocessing, model training, and evaluation) to first- and second-year undergraduate students.

Full-Stack Developer 07/2021 – 09/2022
Smart Friqi Kabul, Afghanistan

- Built ERP modules and administration dashboards using Laravel, Nuxt.js, and PostgreSQL, and added real-time status updates using WebSockets.

Web Developer 05/2021 – 07/2021
TwistedBits ICT Kabul, Afghanistan

- Built two internal management information systems using PHP, JavaScript, Laravel, and Vue.js as part of a small development team.

PROJECTS

Enterprise RAG & Multi-Agent Assistant (Konecranes) 05/2026 – 08/2026
Tech: Python, FastAPI, Next.js, PostgreSQL, AWS Bedrock, LangChain, LangGraph, RAG, AWS (Amplify, ECS, RDS, CloudWatch)

- Internal knowledge assistant over 100+ technical documents (XML, PDF, XLSX). Built a switchable backend to compare three RAG pipelines behind a shared Next.js and FastAPI interface, with multi-agent task routing, PostgreSQL user feedback storage, and AWS deployment.

CRAG Hybrid Retriever — Open-Source Python Retrieval Library 2026
Tech: Python, RAG, Corrective RAG (CRAG), BM25, Dense Vector Search, Reciprocal Rank Fusion (RRF), FlashRank (ONNX), Pydantic, Pytest, GitHub Actions

- Open-source Python library combining BM25 keyword search and dense vector search via Reciprocal Rank Fusion (RRF) so queries match both exact technical codes and semantic meaning without manual weight tuning.
- Added lightweight ONNX cross-encoder reranking (FlashRank), relevance grading to filter out low-confidence passages before LLM generation, IR evaluation metrics (Precision@K, Recall@K, MRR, NDCG@K), and automated GitHub Actions CI (Pytest, Ruff, Mypy).

Freight Forwarding & Logistics ERP System (amirsahra.com) 08/2026 – 10/2026
Tech: PHP, Laravel, Filament, Livewire, PostgreSQL, Tailwind CSS, Spatie PDF, PHPUnit

- Web-based logistics and billing system built for an international freight forwarder, handling Master/House Bills of Lading, container manifests, server-side PDF invoice generation, and balance tracking with PHPUnit test coverage.

Dara.af E-Commerce Platform (dara.af) 05/2024 – 07/2025
Tech: Laravel, PostgreSQL, React, Next.js, REST APIs

- Led an 8-engineer team building a production e-commerce marketplace with product catalogs, order tracking, merchant administration, PostgreSQL schema design, and automated checkout tests.

PGL ERP System (pglsystem.com) 09/2022 – 04/2024
Tech: NestJS, React, Next.js, PostgreSQL, Tailwind CSS, REST APIs

- Internal ERP system managing international shipments, customs clearance, warehouse stock, HR, and finance workflows across company departments.

EDUCATION

Master's Degree in Artificial Intelligence 08/2025 – Present
University of Jyväskylä Jyväskylä, Finland

- Currently in progress (Expected May 2027) · Master's Thesis: Exploring LLM-assisted validation for multi-objective optimization in Finnish forest planning.

Bachelor of Computer Science (Software Engineering) 2017 – 2021
Kabul University Kabul, Afghanistan

LANGUAGE SKILLS

- **English** (Fluent)
- **Persian** (Native)
- **Finnish** (A1, actively learning)